Neyman Orthogonality

Definitions, generalisations and estimators without a score

Jan Teichert-Kluge

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Neyman orthogonality

Data, target and Riesz representer

We observe n independent copies of $W = (Y, Z)$ with law P, where Y is an outcome and Z collects the regressors, and write $g_0(z) = \mathbb{E}[Y \mid Z = z]$ for the outcome regression and $\langle f, g \rangle = \mathbb{E}[f(Z)g(Z)]$ for the inner product of $L^2(P_Z)$.

The target is a linear functional $\theta_0 = \theta(g_0)$ of the outcome regression, with $\theta(g) = \mathbb{E}[\nu(W; g)]$. When θ is mean-square continuous, by the Riesz representation theorem (Riesz 1907; Fréchet 1907) there is a unique $\alpha_0 \in L^2(P_Z)$ with

$$\theta(g) = \langle \alpha_0, g \rangle \quad \text{for all } g \in L^2(P_Z), \qquad \text{hence} \qquad \theta(g_0 + h) - \theta_0 = \langle \alpha_0, h \rangle.$$

The representer α_0 is therefore the derivative of the target with respect to the regression, and $\theta_0 = \mathbb{E}[\alpha_0(Z)\, Y]$ (Chernozhukov, Newey, and Singh 2022, sec. 2).

Target, with $Z = (D, X)$	$\nu(W; g)$	$\alpha_0(Z)$
average treatment effect (ATE), $D \in \{0, 1\}$	$g(1, X) - g(0, X)$	$D/m_0(X) - (1 - D)/\{1 - m_0(X)\}$
coefficient θ_0 in $Y = \theta_0 D + f_0(X) + \varepsilon$, $D = m_0(X) + V$	$V\, g(Z)/\mathbb{E}[V^2]$	$V/\mathbb{E}[V^2]$
weighted average derivative, D continuous	$\int \omega(u)\, \partial_d g(u, X)\, \mathrm{d}u$	$-\omega'(D)/\varrho(D \mid X)$

For the ATE, $m_0(x) = P(D = 1 \mid X = x)$ is the propensity score, and the target is identified under consistency, unconfoundedness and overlap (Rosenbaum and Rubin 1983). The partially linear model is that of Robinson (1988), with $\mathbb{E}[\varepsilon \mid D, X] = 0$ and $\mathbb{E}[V \mid X] = 0$. For the derivative, ω is a known density and $\varrho(d \mid x)$ is the density of D given X (Chernozhukov, Newey, and Singh 2022, Table 1).

Scores, moment conditions and cross-fitting

A score is a map $\psi(w;\theta,\eta)$ with a nuisance η in a convex set T, and $M(\theta,\eta)=\mathbb{E}_P[\psi(W;\theta,\eta)]$ is its moment map. The score identifies θ_0 if $M(\theta_0,\eta_0)=0$, this zero is unique, and the Jacobian $J_0=\partial_\theta M(\theta,\eta_0)|_{\theta=\theta_0}$ is non-singular. Examples are

$$\begin{aligned}
\psi_{\mathrm{DR}}(W;\theta,g,\alpha) &= \nu(W;g)-\theta+\alpha(Z)\{Y-g(Z)\},\\
\psi_{\mathrm{IV}}(W;\theta,f,m) &= \{Y-\theta D-f(X)\}\{D-m(X)\},\\
\psi_{\mathrm{PO}}(W;\theta,\ell,m) &= \{Y-\ell(X)-\theta(D-m(X))\}\{D-m(X)\},
\end{aligned}$$

where ψ_{DR} is the augmented inverse probability weighting (AIPW) score of any linear functional (J. M. Robins, Rotnitzky, and Zhao 1994; Chernozhukov, Newey, and Singh 2022, eq. 2.4), and ψ_{IV} and ψ_{PO} are the scores of the partially linear model with $\ell_0(X)=\mathbb{E}[Y\mid X]=\theta_0 m_0(X)+f_0(X)$ (Chernozhukov et al. 2018, eqs. 4.3–4.4).

A conditional moment restriction $\mathbb{E}[\rho(W;\theta_0,\eta_0)\mid Z]=0$ implies $\mathbb{E}[A(Z)\,\rho(W;\theta_0,\eta_0)]=0$ for every instrument $A(Z)$ (Chamberlain 1987). The sensitivity of the moment condition to the nuisance and the efficiency of the estimator depend on the instrument.

For the double machine learning (DML) estimator, we partition the sample into folds $I_1,\dots,I_K$, compute $\hat\eta_k$ from the observations outside I_k, and solve (Chernozhukov et al. 2018, Def. 3.2)

$$\frac{1}{n}\sum_{k=1}^{K}\sum_{i\in I_k}\psi(W_i;\theta,\hat\eta_k)=0.$$

Three-term decomposition

With $\mathbb{E}_{n,k}$ the average over fold I_k, $\mathbb{P}_n$ the average over the whole sample and $M(\theta_0,\eta_0)=0$, the average of the score splits as (Kennedy 2024)

$$\mathbb{E}_{n,k}\psi(W;\theta_0,\hat\eta_k)=\underbrace{\mathbb{E}_{n,k}\psi(W;\theta_0,\eta_0)}_{T_1}+\underbrace{(\mathbb{E}_{n,k}-P)\{\psi(\cdot;\theta_0,\hat\eta_k)-\psi(\cdot;\theta_0,\eta_0)\}}_{T_2}+\underbrace{M(\theta_0,\hat\eta_k)}_{T_3},$$

and along the segment from η_0 to $\hat\eta_k$,

$$T_3=\partial_r M(\theta_0,\eta_0+r(\hat\eta_k-\eta_0))\big|_{r=0}+\int_0^1(1-r)\,\partial_r^2 M(\theta_0,\eta_0+r(\hat\eta_k-\eta_0))\,\mathrm{d}r.$$

Term	Order	Made negligible by
T_1	$O_P(n^{-1/2})$ and asymptotically normal	central limit theorem, which for the ATE requires overlap

Term	Order	Made negligible by
T_2	centred average of n/K terms given $\hat{\eta}_k$	cross-fitting with consistent $\hat{\eta}_k$, or stability of the nuisance estimator
T_3, first part	linear in $\hat{\eta}_k - \eta_0$	Neyman orthogonality of the score
T_3, second part	quadratic in $\hat{\eta}_k - \eta_0$	rates of convergence of $\hat{\eta}_k$

For the plug-in score $\nu(W; g) - \theta$, $T_3 = \langle \alpha_0, \hat{g}_k - g_0 \rangle$ exactly. Nonparametric estimators of g_0 converge more slowly than $n^{-1/2}$, so $\sqrt{n}(\hat{\theta} - \theta_0)$ diverges unless this term is $o_P(n^{-1/2})$.

Definition

Following Chernozhukov et al. (2018, sec. 2.1), let $\widetilde{T} = \{\eta - \eta_0 : \eta \in T\}$ and define the Gateaux derivative map $\mathrm{D}_r : \widetilde{T} \to \mathbb{R}^{d_\theta}$ by

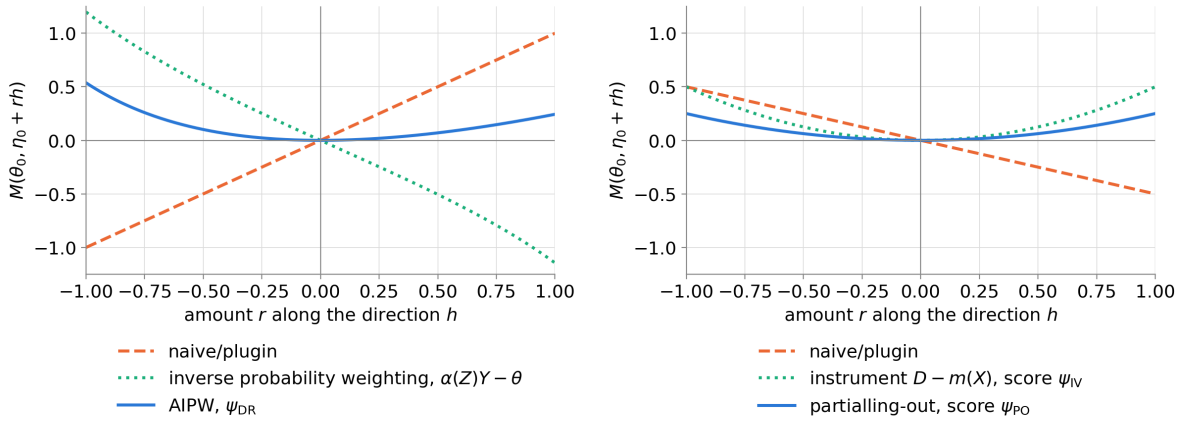
$$\mathrm{D}_r[\eta - \eta_0] = \partial_r \Big\{ \mathbb{E}_P[\psi(W; \theta_0, \eta_0 + r(\eta - \eta_0))] \Big\}, \qquad r \in [0, 1).$$

Definition (Chernozhukov et al. 2018, Def. 2.1). The score ψ obeys the Neyman orthogonality condition at (θ_0, η_0) with respect to the nuisance realisation set $\mathcal{T}_N \subset T$ if $M(\theta_0, \eta_0) = 0$, the map $\mathrm{D}_r[\eta - \eta_0]$ exists for all $r \in [0, 1)$ and $\eta \in \mathcal{T}_N$, and

$$\partial_\eta \mathbb{E}_P \psi(W; \theta_0, \eta_0)[\eta - \eta_0] := \mathrm{D}_0[\eta - \eta_0] = 0 \qquad \text{for all } \eta \in \mathcal{T}_N.$$

- The derivative is taken at the truth with the target held at θ_0. It is the derivative of the moment map, which equals $\mathbb{E}_P \partial_\eta \psi$ only when differentiation under the integral is allowed.
- The set $\mathcal{T}_N$ contains the nuisance estimates with high probability, where N is the sample size, and a smaller set imposes a weaker condition. The condition therefore depends on the law P and on the set $\mathcal{T}_N$ as well as on the score.
- The score obeys the λ_N near-orthogonality condition if $\|\mathrm{D}_0[\eta - \eta_0]\| \leq \lambda_N = o(N^{-1/2})$ for all $\eta \in \mathcal{T}_N$ (Chernozhukov et al. 2018, Def. 2.2), which arises when the parameter that orthogonalises the score is defined by a regularised problem.

Moment maps along a nuisance direction



(a) ATE with a binary treatment

(b) Partially linear model

With X uniform on $[0, 1]$ and $\theta_0 = 1$, panel (a) uses $m_0(x) = 0.25 + 0.5x$, $g_0(d, x) = d + 2x$ and the direction $(g, m) = (g_0 + rd,\ m_0 + 0.15r)$, and panel (b) uses $m_0(x) = x$, $f_0(x) = x^2$ and the directions $f_0 + r$, $\ell_0 + r$ and $m_0 + 0.5r$. The slope at $r = 0$ is $\mathrm{D}_0[h]$, which vanishes for the AIPW score and for both scores of the partially linear model.

Construction of orthogonal scores

Correction by the Riesz representer. For a fixed $A \in L^2(P_Z)$, the score $\nu(W; g) - \theta + A(Z)\{Y - g(Z)\}$ has the derivative $\mathrm{D}_0[h] = \mathbb{E}[\nu(W; h)] - \mathbb{E}[A(Z)h(Z)] = \langle \alpha_0 - A,\ h \rangle$ in a direction h of g. It is orthogonal in g for a dense set of directions exactly when $A = \alpha_0$, so the correction term of an orthogonal score of a linear functional is the Riesz representer times the residual. With α as a second nuisance, the score is orthogonal in α because $\mathbb{E}[Y - g_0(Z) \mid Z] = 0$.

Orthogonal instruments. In the partially linear model, the moment $\mathbb{E}[A(D, X)\{Y - \theta_0 D - f(X)\}]$ has the derivative $-\mathbb{E}[\mathbb{E}[A \mid X]\, h(X)]$ in a direction $h(X)$ of f, so it is orthogonal in f exactly when $\mathbb{E}[A \mid X] = 0$. Every instrument $\{D - m_0(X)\}k(X)$ satisfies this condition, so orthogonal scores are not unique, and the choice of k affects the efficiency (Chernozhukov et al. 2018, sec. 2.2.4).

Projection on the nuisance scores. In a regular parametric model with scores s_θ and s_η and information matrix with blocks $I_{\theta\eta}$ and $I_{\eta\eta}$, let $\mathbb{E}_{\theta_0, \eta}\psi(W; \theta_0, \eta) = 0$ for all η. Differentiating in η gives $\partial_\eta M(\theta_0, \eta_0) = -\mathbb{E}[\psi_0 s_\eta^\top]$, where $\psi_0 = \psi(\cdot; \theta_0, \eta_0)$. The residual score of Neyman (1959, 1979),

$$\psi = s_\theta - I_{\theta\eta} I_{\eta\eta}^{-1} s_\eta,$$

is uncorrelated with s_η and therefore orthogonal. Chernozhukov et al. (2018, sec. 2.2) construct orthogonal scores in likelihood and moment problems by projecting a score off the nuisance directions, and the orthogonalising matrix becomes a further nuisance.

Double robustness and the mixed bias property

Let the nuisance be a pair $(g, \alpha) \in \mathcal{T}_g \times \mathcal{T}_\alpha$. The score is doubly robust if $M(\theta_0, g, \alpha_0) = M(\theta_0, g_0, \alpha) = 0$ for all g and α, and it has the mixed bias property if $M(\theta_0, g, \alpha) = \mathbb{E}[S(W)\,(g - g_0)(Z)\,(\alpha - \alpha_0)(Z)]$ for all (g, α) and a known function S (A. Rotnitzky, Smucler, and Robins 2021). The AIPW score has the mixed bias property with $S = -1$ (Chernozhukov, Newey, and Singh 2022, eq. 2.5),

$$M(\theta_0, g, \alpha) = -\langle g - g_0,\, \alpha - \alpha_0 \rangle \qquad \text{for every pair } (g, \alpha).$$

Proposition. Let $T = \mathcal{T}_g \times \mathcal{T}_\alpha$ be a product of convex sets, let $\mathcal{T}_N = T$, and let D_0 be linear. Then the following hold.

1. The mixed bias property implies double robustness, and double robustness implies orthogonality.
2. If $M(\theta_0, g_0, \alpha) = 0$ for every α, the score is doubly robust exactly when it is orthogonal and $g \mapsto M(\theta_0, g, \alpha_0)$ is affine (Chernozhukov, Escanciano, et al. 2022, Thm. 4).
3. Orthogonality does not imply double robustness. In the partially linear model,

   $$M_{\mathrm{PO}}(\theta_0, \ell, m) = \langle \ell - \ell_0,\, m - m_0 \rangle - \theta_0 \|m - m_0\|^2, \qquad M_{\mathrm{IV}}(\theta_0, f, m) = \langle f - f_0,\, m - m_0 \rangle,$$

   so both scores are orthogonal, and for $\theta_0 \neq 0$ only ψ_{IV} is doubly robust.

Since $\ell_0 = \theta_0 m_0 + f_0$ contains m_0, an error in m also enters through ℓ. The choice of the nuisance that is called the outcome model matters for double robustness and does not matter for orthogonality.

Properties and limits of orthogonal scores

Properties

- The nuisance error enters the bias term T_3 only through second-order terms, which are $o_P(n^{-1/2})$ when $\|\hat{\eta}_k - \eta_0\| = o_P(n^{-1/4})$ in $L^2(P)$, or, for scores with the mixed bias property, when $\|\hat{g} - g_0\|\,\|\hat{\alpha} - \alpha_0\| = o_P(n^{-1/2})$, so any learner with these rates can estimate the nuisance.
- With cross-fitting and these rates, the DML estimator is asymptotically linear with influence function $-J_0^{-1}\psi_0$ without Donsker conditions on the nuisance class, and Wald intervals with a plug-in variance estimator are valid uniformly over models in which the rates hold uniformly (Chernozhukov et al. 2018, Thm. 3.1, Thm. 3.2 and Cor. 3.1).

- If an orthogonal score identifies the target along a nonparametric model, $-J_0^{-1}\psi_0$ is the efficient influence function (EIF), the influence function of an efficient estimator, and the DML estimator attains the efficiency bound (Appendix).
- Under the mixed bias property the bias formula holds for every nuisance pair, so nuisance errors from any source enter through one inner product.

Limits

- Orthogonality removes only the first-order part of T_3, and nuisance errors that do not shrink with n violate the rate conditions.
- Overlap is needed for $\alpha_0 \in L^2(P_Z)$ and for a bounded $\hat{\alpha}$, and trimming changes the estimand.
- Identification of a causal target requires assumptions about the process that generates the data, which do not depend on the score.
- Among the orthogonal scores the EIF has the smallest variance, and ψ_{PO} is efficient when $\mathrm{Var}(\varepsilon \mid D, X)$ is constant.
- The definition applies to estimators that solve an estimating equation.

Generalisations of Neyman orthogonality

Approaches by term of the decomposition

Approach	Term	Representative references
correction of the score	first part of T_3	(Newey 1994; Chernozhukov et al. 2018; Chernozhukov, Escanciano, et al. 2022)
automatic or numerical correction	first part of T_3	(Chernozhukov, Newey, and Singh 2022; Chernozhukov, Newey, Quintas-Martínez, et al. 2022; Carone, Luedtke, and Laan 2019; Ichimura and Newey 2022)
fit that solves the orthogonal equation	first part of T_3	(Scharfstein, Rotnitzky, and Robins 1999; M. J. van der Laan and Rubin 2006; Bruns-Smith et al. 2026)
balancing and minimax weights	first part of T_3	(Zubizarreta 2015; Kallus 2020; Hirshberg and Wager 2021)
prior invariant along the least-favourable direction	first part of T_3	(Castillo and Rousseau 2015; Ray and van der Vaart 2020; Breunig, Liu, and Yu 2025; Melnychuk, Javurek, and Feuerriegel 2026)
higher-order orthogonality and influence functions	higher-order parts of T_3	(Mackey, Syrgkanis, and Zadik 2018; J. Robins et al. 2008; J. M. Robins et al. 2017)
cross-fitting, stability of the nuisance estimator	T_2	(Chernozhukov et al. 2018; Chen, Syrgkanis, and Austern 2022)

Approach	Term	Representative references
calibration and further targeting	second part of T_3, one slow nuisance	(Benkeser et al. 2017; L. van der Laan, Luedtke, and Carone 2024)

Automatic debiased machine learning

The correction term of an orthogonal score of a linear functional contains the Riesz representer, which minimises the Riesz loss

$$\mathbb{E}[\alpha(Z)^2 - 2\nu(W;\alpha)] = \|\alpha - \alpha_0\|^2 - \|\alpha_0\|^2.$$

The loss can be evaluated without a formula for α_0 when ν does not depend on P, as for the ATE, and minimising an empirical version of it over a class of functions is called Riesz regression (Chernozhukov et al. 2021).

- Chernozhukov, Newey, and Singh (2022) minimise a Lasso version of the Riesz loss over linear α, use the fit in the cross-fitted AIPW score of any mean-square continuous linear functional, and call the method automatic debiased machine learning.
- RieszNet learns g and α with a neural network that shares its layers between the two, and ForestRiesz learns α with a random forest (Chernozhukov, Newey, Quintas-Martínez, et al. 2022).
- Chernozhukov et al. (2026) give a textbook account of automatic debiased machine learning, and L. van der Laan et al. (2025) extend it to smooth functionals of M-estimands, which are population risk minimisers such as counterfactual regressions, survival functions and conditional average treatment effects, using the gradient and Hessian of the loss in place of an analytic influence function.
- For targets without a known form of the influence function, the correction can be computed numerically, by finite differences of $P \mapsto \theta(P)$ at an estimate of P perturbed toward a smoothed point mass (Carone, Luedtke, and Laan 2019; Jordan, Wang, and Zhou 2022), or as the derivative of the limit of the estimator along contamination paths (Ichimura and Newey 2022).

Orthogonality without an explicit correction

- Targeting. The clever covariate (Scharfstein, Rotnitzky, and Robins 1999; Bang and Robins 2005) and targeted maximum likelihood (M. J. van der Laan and Rubin 2006; M. J. van der Laan and Rose 2011) add $\hat{\alpha}$ to the outcome fit so that $\mathbb{P}_n[\hat{\alpha}\{Y - \hat{g}\}] = 0$, and the plug-in at the modified fit equals the corrected estimator.

- Sieves. Least squares on a closed subspace $\mathcal{S}$, with projection $\mathsf{P}_{\mathcal{S}}$ and $\mathsf{M}_{\mathcal{S}} = I - \mathsf{P}_{\mathcal{S}}$, solves the orthogonal equation for $\mathsf{P}_{\mathcal{S}}\alpha_0$, and the plug-in bias is $-\langle \mathsf{M}_{\mathcal{S}}\alpha_0, \mathsf{M}_{\mathcal{S}}g_0 \rangle$ (Chernozhukov, Newey, and Singh 2022, eq. 2.6). Sieve plug-ins can be efficient for smooth functionals (Newey 1994; Shen 1997).
- Balancing weights. Weighting estimators $n^{-1}\sum_i w_i Y_i$ can be designed so that the weights balance a class of outcome functions, exactly or up to a bound on the worst-case imbalance (Zubizarreta 2015; Kallus 2020; Hirshberg and Wager 2021). When the outcome and weighting models are linear in the same basis, augmented balancing weights are equivalent to a single linear regression (Bruns-Smith et al. 2026).
- Priors. The no-bias condition of a semiparametric Bernstein–von Mises theorem (Castillo and Rousseau 2015) holds when the prior is approximately invariant under shifts of order $n^{-1/2}$ along the least-favourable direction, which for the ATE is proportional to the Riesz representer when $\mathrm{Var}(Y \mid Z)$ is constant (Ray and Szabó 2019; Breunig, Liu, and Yu 2025). Priors that depend on an estimated propensity score achieve this invariance (Ray and van der Vaart 2020). Javurek et al. (2026) put a prior on the causal target and update it with a loss that identifies the target in a generalised posterior, which for Neyman-orthogonal losses converges to its oracle counterpart. Other approaches correct the posterior (Breunig, Liu, and Yu 2025; Yiu et al. 2025) or change the outcome model, by a reparametrisation (P. R. Hahn et al. 2018) or with the estimated propensity score as a covariate (P. R. Hahn, Murray, and Carvalho 2020).

Learned representations and in-context nuisance learners

- Neural networks for treatment effects learn representations that predict both the treatment and the outcome, as DragonNet with its targeted regularisation (Shi, Blei, and Veitch 2019) and the adapted text embeddings of Veitch, Sridhar, and Blei (2020).
- Melnychuk, Frauen, et al. (2026) argue that end-to-end representation learning lacks the quasi-oracle efficiency of Neyman-orthogonal learners, and they propose OR-learners, which combine learned representations with Neyman-orthogonal learners of causal quantities.
- TabPFN, a prior-data fitted network (PFN) for tabular prediction (Hollmann et al. 2025), has been used as nuisance learner in meta-learners for heterogeneous effects (Q. Zhang et al. 2025) and in the DoubleML software (Bach et al. 2024).
- Y. Wang, Secrest, and Tchakoute (2026) add a propensity output to CausalPFN, trained jointly with its outcome model, and build AIPW and TMLE estimators on it. These estimators remain stable as confounding grows, whereas the Bayesian plug-in estimate of CausalPFN degrades.
- Optimal covariate balancing is dual to a regularised self-attention, and J. Zhang et al. (2024) learn balancing weights across data sets.
- For a simplified transformer, the variance of a PFN vanishes as n grows while its bias need not (Nagler 2023).

Causal prior-data fitted networks

- Causal PFNs, such as Do-PFN (Robertson et al. 2025), CausalPFN (Balazadeh et al. 2025) and CausalFM (Ma et al. 2026), return effect estimates from the data set in context, without an orthogonal score.
- In simulations based on clinical data, the estimates of CausalPFN were biased and its credible intervals covered poorly (Mourão et al. 2026).
- H. Wang et al. (2026) prove that one structural causal model induces an exponentially large space of interventional distributions, so that pretraining leaves part of them uncovered, which leads to posterior inconsistency and estimation bias, and they propose fine-tuning for specific interventions.
- Melnychuk, Balazadeh, et al. (2026) show that PFNs used as Bayesian ATE estimators have a confounding bias induced by the prior, which they identify with regularisation-induced confounding, the bias of a regularised outcome regression that shrinks the coefficients of covariates that also predict the treatment (P. R. Hahn et al. 2018), and they correct it by a one-step correction of martingale posteriors of TabPFN.
- Z. Zhang (2026) pretrains causal tabular models on the labels $\theta(P) + \mathbb{P}_n\varphi_P$, the ATE plus the fluctuation term of its EIF, and derives the coverage of Wald intervals for the frozen network.

Neyman orthogonality is defined for scores, and causal PFNs that return the effect directly have none.

Debiased prior-data fitted networks

Melnychuk, Javurek, and Feuerriegel (2026) train a PFN with the log loss on the target θ_ξ of data-generating processes ξ drawn from a prior Π. They call the network idealised when it returns the posterior of θ_ξ exactly, and they study this posterior under data from a fixed law P for general targets with an EIF, including the ATE.

- Without a correction, the L^2 concentration error of the target posterior is at most the posterior average of the nuisance error plus a second-order remainder, and a worst-case posterior inherits the nuisance rate (Melnychuk, Javurek, and Feuerriegel 2026, Thm. 1).
- Label correction. The network is trained on the labels $\theta_\xi + \sum_i \varpi_i \varphi_\xi(W_i)$, with the EIF φ_ξ under ξ and Bayesian bootstrap weights $(\varpi_1, \dots, \varpi_n) \sim \mathrm{Dirichlet}(1, \dots, 1)$ drawn independently of ξ and $\mathcal{D}_n$, so that the one-step posterior correction of Yiu et al. (2025) is applied in pretraining (Melnychuk, Javurek, and Feuerriegel 2026, Eq. 12).
- Prior correction. Each process ξ is perturbed along a least-favourable submodel, a one-dimensional submodel through ξ whose score is φ_ξ, with a coefficient $\kappa \sim N(0, \sigma_n^2)$ and $n\sigma_n^2 \to \infty$, which extends the prior correction of Ray and van der Vaart (2020; Melnychuk, Javurek, and Feuerriegel 2026, Eq. 15). For the ATE with Gaussian noise

of unit variance, the outcome regression g_ξ of ξ is replaced by $g_\xi + \kappa\alpha_\xi$, with the Riesz representer α_ξ of the ATE under ξ, and the density of X is reweighted with the same κ (App. F.2.1).

- With either correction and a stochastically equicontinuous posterior, the posterior of the idealised network concentrates at the rate $n^{-1/2}$ plus the remainder (Thm. 2). When the remainder is $o_P(n^{-1/2})$, a Bernstein–von Mises theorem with the efficient variance holds under regularity conditions (Thm. 3). When the mean square of the remainder is bounded uniformly over a class of laws supported by the prior, the worst-case risk of the posterior mean over this class is of order n^{-1} plus this bound, the minimax rate when a matching lower bound holds (Thm. 4).
- In ATE experiments, the training prior draws the propensity score independently of the outcome regressions, and an evaluation prior has more observed confounding. Under the evaluation prior, the authors report that both corrected networks often have a smaller L^2 risk than the network without a correction for $n \geq 1024$ with 5 covariates and $n \geq 2048$ with 10, and the total variation distance of all networks to the efficient normal approximation grows with n (Sec. 5).

Orthogonality for estimators without a score

Estimators without a score

The following estimators map the data set $\mathcal{D}_n = (W_1, \ldots, W_n)$ to an estimate without an estimating equation.

- Posterior means, which are Bayes rules under a prior.
- Plug-in estimators with regularised or learned outcome regressions, which are subject to regularisation-induced confounding.
- Networks trained end to end on the target, and plug-in estimators on learned representations.
- In-context learners, which are fixed maps F_χ with parameters χ learned in pretraining on synthetic data sets from a prior Π over data-generating processes ξ (Müller et al. 2022).

Lemma. Over all measurable maps F, the squared pretraining loss $\mathbb{E}_{\xi,\mathcal{D}_n}[\{F(\mathcal{D}_n) - \theta_\xi\}^2]$ is minimised, uniquely up to almost sure equality, by the posterior mean $\hat{\theta}_\Pi(\mathcal{D}_n) = \mathbb{E}_\Pi[\theta_\xi \mid \mathcal{D}_n]$ (Müller et al. 2022; Foong et al. 2020).

A trained network computes $\hat{\theta}_\chi = \hat{\theta}_\Pi + a_\chi(\mathcal{D}_n)$, where a_χ is the amortisation error. By the lemma, the minimiser of the pretraining loss is a Bayes rule, whose properties depend on the prior and on the training target. The amortisation error of a fixed network is not guaranteed to vanish as n grows, so rate conditions on in-context nuisances cannot be verified from the pretraining objective alone.

First-order insensitivity

Outcome shifts. For a direction $h \in L^2(P_Z)$ and $\delta \in \mathbb{R}$, let $P_{\delta h}$ be the law of $(Y + \delta h(Z), Z)$. Under $P_{\delta h}$ the outcome regression is $g_0 + \delta h$ and the target is $\theta_0 + \delta\langle\alpha_0, h\rangle$. Write $b_n(\delta; h) = \mathbb{E}_{\delta h}[\hat{\theta}] - \theta(P_{\delta h})$ for the bias of an estimator $\hat{\theta}$ along the shift.

Proposed definition. The estimator $\hat{\theta}$ is first-order insensitive to the outcome regression in the direction h if $\partial_\delta b_n(0; h) = 0$, and asymptotically first-order insensitive over a class of directions if this derivative tends to zero uniformly over the class.

- The bias then does not drift at the scale $n^{-1/2}$ along local shifts. The condition is the derivative part of local asymptotic unbiasedness (Hirano and Porter 2012) and, under uniform integrability, equivalent to regularity along these shifts (van der Vaart 1998, Ch. 8).
- The definition needs only the map from data sets to estimates, so it applies to trained networks, whereas Melnychuk, Javurek, and Feuerriegel (2026) analyse the idealised network.

Causal PFNs under independent mechanisms

Write a task as $\xi = (\xi_X, \xi_D, \xi_Y)$ with $P_\xi = P_{X;\xi_X} \otimes P_{D|X;\xi_D} \otimes P_{Y|D,X;\xi_Y}$. The prior Π has independent mechanisms if ξ_D is independent of (ξ_X, ξ_Y) under Π.

Proposition. Under independent mechanisms, $\Pi(\mathrm{d}\xi_X, \mathrm{d}\xi_Y \mid \mathcal{D}_n) \propto \Pi_{X,Y}(\mathrm{d}\xi_X, \mathrm{d}\xi_Y) \prod_i p(X_i; \xi_X)\, p(Y_i \mid D_i, X_i; \xi_Y)$ does not involve the treatment likelihood, so the posterior mean of the ATE is a Bayesian outcome regression with the uncorrected term $\langle\alpha_0, \hat{g}_\Pi - g_0\rangle$ in its error, where $\hat{g}_\Pi = \mathbb{E}_\Pi[g_\xi \mid \mathcal{D}_n]$. The factorisation is the Bayesian ignorability of the treatment assignment (Gelman et al. 2013, sec. 8.2; Li, Ding, and Mealli 2023, sec. 3).

Model	Treatment mechanism	Outcome mechanism	Prior dependence
Do-PFN (Robertson et al. 2025)	node of a random structural causal model	node of the same graph	shared graph, independent weights
CausalPFN (Balazadeh et al. 2025)	logit of X that is constant, linear or a random network	relabelled columns of a base table	independent draws
CausalFM (Ma et al. 2026)	random network $f_A(X, U_A)$	random network $f_Y(X, A, U_Y)$	independent weights and noise

Proposed result. Under independent mechanisms, the Bayes rule for a training target that is a function of $(\xi_X, \xi_Y, \mathcal{D}_n)$ does not involve the treatment likelihood. The minimiser of the squared pretraining loss therefore involves the treatment likelihood only when the training

target depends on ξ_D or the prior couples ξ_D with (ξ_X, ξ_Y), as the priors of Li, Ding, and Mealli (2023, sec. 5) do, and the label correction and the prior correction of Melnychuk, Javurek, and Feuerriegel (2026) are instances of the two cases.

Project question and motivation

Motivation. Causal PFNs return an effect estimate in one forward pass, but they have no score, so Neyman orthogonality cannot be checked for them. In simulations based on clinical data, CausalPFN was biased and its credible intervals covered poorly (Mourão et al. 2026). The guarantees of Melnychuk, Javurek, and Feuerriegel (2026) hold for the idealised network, and their theory sets the amortisation error of a trained network aside.

Main question. When is an estimator without a score, in particular a trained in-context learner, first-order insensitive to errors in the outcome regression, and how can this be checked from the data and the trained network alone?

1. Which property of an estimator takes the place of Neyman orthogonality, how is it related to regularity and to orthogonal scores, and which shifts beyond those of the outcome regression does it have to cover?
2. Do the implicit weights $w_i = n\,\partial\hat{\theta}/\partial Y_i$ of a trained network give a diagnostic of this property, and under which conditions does a test based on it have a known level?
3. Which training targets and priors make the Bayes rule of an in-context learner first-order insensitive?
4. Are trained networks first-order insensitive when they use the corrections of Melnychuk, Javurek, and Feuerriegel (2026), or in-context nuisances with cross-fitting and an AIPW layer?

Core contribution and positioning

Proposed contribution	Closest work	What is new
A criterion without a score, first-order insensitivity along shifts of the outcome regression	local asymptotic unbiasedness (Hirano and Porter 2012); Giordano et al. (2026)	the derivative condition of local asymptotic unbiasedness at fixed n, written for every estimator differentiable in the outcomes as balance of its implicit weights, with the Riesz representer as target
A diagnostic from one backward pass, the imbalance of the implicit weights	balance checks of linear weights (Chattopadhyay and Zubizarreta 2023; Knaus 2024); Giordano et al. (2026)	estimators other than posterior means, such as trained networks, the expected imbalance as the sensitivity, and a test with a known level

Proposed contribution	Closest work	What is new
Bayes rules of in-context learners under the label and prior corrections	Melnychuk, Javurek, and Feuerriegel (2026); Z. Zhang (2026); Li, Ding, and Mealli (2023); Linero (2021)	when the treatment mechanism enters the Bayes rule, the AIPW form of the Bayes rule of the label correction, and finite-sample unbiasedness of a prior correction in a Gaussian model
Measurement of the implicit weights of trained causal PFNs and of designs with in-context nuisances, such as an AIPW layer on nuisance heads and cross-fitting in context	Mourão et al. (2026); Y. Wang, Secrest, and Tchakoute (2026); Melnychuk, Javurek, and Feuerriegel (2026)	first-order insensitivity of trained networks, measured with the diagnostic

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Appendix

Asymptotic normality of the DML estimator

Let $\psi = \psi^a(W;\eta)\,\theta + \psi^b(W;\eta)$ be Neyman orthogonal with respect to $\mathcal{T}_N$, and let $\hat{\eta}_k \in \mathcal{T}_N$ with probability at least $1-\Delta_N$, where $\Delta_N \to 0$, $\delta_N \to 0$ and

$$r'_N = \sup_{\eta\in\mathcal{T}_N} \left(\mathbb{E}_P\|\psi(W;\theta_0,\eta) - \psi(W;\theta_0,\eta_0)\|^2\right)^{1/2} \le \delta_N,$$
$$\lambda'_N = \sup_{r\in(0,1),\,\eta\in\mathcal{T}_N} \left\|\partial_r^2 M(\theta_0, \eta_0 + r(\eta-\eta_0))\right\| \le \delta_N N^{-1/2}.$$

Under these conditions and the moment and identification conditions of Assumptions 3.1 and 3.2 of Chernozhukov et al. (2018), by their Theorem 3.1,

$$\sqrt{N}(\hat{\theta}-\theta_0) = -J_0^{-1}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\psi(W_i;\theta_0,\eta_0)+o_P(1) \;\rightsquigarrow\; N(0,\sigma^2), \qquad \sigma^2 = J_0^{-1}\,\mathbb{E}_P[\psi_0\psi_0^\top]\,(J_0^{-1})^\top,$$

uniformly over the laws that satisfy the assumptions, and Wald intervals with the plug-in variance estimator are uniformly valid (Chernozhukov et al. 2018, Thm. 3.2 and Cor. 3.1).

By the bound on r'_N and cross-fitting, $T_2 = o_P(N^{-1/2})$. By the bound on λ'_N, the second part of T_3 is $o_P(N^{-1/2})$, which for smooth scores holds when the L^2 rate ε_N of the nuisance estimator is $o(N^{-1/4})$, and for ψ_{DR}, where $\partial_r^2 M = -2\langle g-g_0, \alpha-\alpha_0\rangle$, when the product of the two errors is $o(N^{-1/2})$.

Orthogonal scores and efficient influence functions

A regular submodel is a path $t \mapsto P_t$ through P, differentiable in quadratic mean, whose log density has derivative s at $t = 0$, and $L_0^2(P)$ is the space of square-integrable functions with mean zero. A function $\varphi \in L_0^2(P)$ is a gradient of θ at P if $\partial_t \theta(P_t)|_{t=0} = \mathbb{E}[\varphi\, s]$ for every regular submodel, and in a locally nonparametric model, in which these derivatives s are dense in $L_0^2(P)$, the gradient is unique and is the efficient influence function (EIF) (van der Vaart 1998, Ch. 25). If ψ identifies θ along a regular submodel, $\mathbb{E}_{P_t}\psi(W; \theta(P_t), \eta(P_t)) = 0$, then differentiation at $t = 0$ gives

$$0 = \mathbb{E}_P[\psi_0\, s] + J_0\, \dot{\theta} + \mathrm{D}_0[\dot{\eta}],$$

with $\dot{\theta}$ and $\dot{\eta}$ the derivatives of $\theta(P_t)$ and $\eta(P_t)$ at $t = 0$. Hence D_0 vanishes on all nuisance tangents $\dot{\eta}$ exactly when $-J_0^{-1}\psi_0$ is a gradient, and in a locally nonparametric model a score that identifies θ along the model is orthogonal exactly when $-J_0^{-1}\psi_0$ is the EIF (Newey 1994; Chernozhukov, Escanciano, et al. 2022). For the ATE, $\psi_{\mathrm{DR}}(\cdot; \theta_0, g_0, \alpha_0)$ is the EIF (J. Hahn 1998).

Formulations of orthogonality

The formulations below are equivalent to the definition, or sufficient for it, under conditions on the model and the directions.

Formulation	Condition	Source
conditional form	the derivative of the score in the value of the nuisance at the data point has conditional mean zero	Mackey, Syrgkanis, and Zadik (2018)
local robustness, no first-step adjustment	the derivative of the moment along mixtures of P with another law vanishes	Newey (1994); Chernozhukov, Escanciano, et al. (2022)
nuisance tangent space	ψ_0 is orthogonal in $L^2(P)$ to the closed span of the derivatives s of submodels that move only η	Tsiatis (2006)
parameter orthogonality	the likelihood score of θ is uncorrelated with that of η	Cox and Reid (1987)
orthogonal loss	the cross derivative of the population risk in the target and the nuisance vanishes at the minimiser	Foster and Syrgkanis (2023)

Local robustness and general targets

- Newey (1994) derives the adjustment term that a nonparametric first step adds to the influence function of a two-step estimator, and the term vanishes for orthogonal scores.
- Chernozhukov, Escanciano, et al. (2022) construct locally robust moments $\psi = q + \phi$ for the generalised method of moments and for nonlinear functionals, where ϕ is the influence function of the first step along paths $F_\tau = (1-\tau)F_0 + \tau H$.
- Chernozhukov, Newey, Singh, et al. (2022) give debiased estimators for dynamic treatment effects and general nested functionals, in which each regression is a nuisance of the next.
- Bennett et al. (2026) study functionals that are strongly identified while the nuisance is weakly identified, as in instrumental variable and proxy models. A representer in the range of the adjoint of the operator that defines the nuisance is necessary for $\sqrt{n}$-estimation.
- Foster and Syrgkanis (2023) extend orthogonality to population risks $L(\theta, g)$ whose minimiser θ is a function, such as a conditional average treatment effect, and the excess risk of the second stage is then insensitive to first order to the nuisance error. The R-learner loss $\mathbb{E}[\{Y - \ell(X) - \tau(X)(D - m(X))\}^2]$ of Nie and Wager (2021) is orthogonal in (ℓ, m).

Higher-order terms and minimax rates

- Mackey, Syrgkanis, and Zadik (2018) call a moment k-orthogonal if its conditional derivatives in the nuisance vanish up to order k, and then a nuisance rate of $o(n^{-1/(2k+2)})$ in the $L^{2(k+1)}$ norm suffices (Mackey, Syrgkanis, and Zadik 2018, Cor. 4). In the partially linear model with conditionally Gaussian treatment noise, every twice differentiable 2-orthogonal score has $J_0 = 0$ and does not identify θ_0 (Mackey, Syrgkanis, and Zadik 2018, Thm. 5). Bonhomme et al. (2026) construct moment functions that are Neyman orthogonal to a chosen order in parametric moment-condition models, with a number of additional nuisance parameters that does not grow with the order.
- Higher-order influence functions estimate higher-order terms of the von Mises expansion by U-statistics and attain minimax rates for nuisance functions too rough for $\sqrt{n}$-estimation (J. Robins et al. 2008; J. M. Robins et al. 2017).
- When the nuisance functions satisfy no assumptions beyond the L^2 errors of their estimates, the product rate of first-order estimators is minimax optimal for quadratic functionals such as the expected conditional covariance (Balakrishnan, Kennedy, and Wasserman 2026), and the product rate of doubly robust estimators is minimax optimal for the ATE (Jin and Syrgkanis 2024).

Forms of robustness and inference

- A. Rotnitzky, Smucler, and Robins (2021) characterise the parameters whose influence functions have the mixed bias property, and Smucler, Rotnitzky, and Robins (2019) distinguish rate double robustness, asymptotic normality whenever both nuisance estimators are consistent and $\|\hat{g} - g_0\| \, \|\hat{\alpha} - \alpha_0\| = o_P(n^{-1/2})$, from model double robustness, the same conclusion when one nuisance estimator converges to the truth and the other to a different limit.
- With more than two nuisances, robustness takes the form of multiple robustness in mediation analysis (Tchetgen Tchetgen and Shpitser 2012) and of sequential double robustness in longitudinal models, where the bias is a sum of products of errors, one per stage (Luedtke et al. 2017; Andrea Rotnitzky, Robins, and Babino 2017).
- A doubly robust estimator can be made asymptotically linear when only one nuisance is estimated consistently, by further targeting (Benkeser et al. 2017) or by calibration (L. van der Laan, Luedtke, and Carone 2024), and leave-one-out stability of the nuisance estimator can replace sample splitting (Chen, Syrgkanis, and Austern 2022).
- Estimators that use working models selected from the data are not regular, so inference for θ_0 is not uniformly valid over the full model even when the score is orthogonal (L. van der Laan et al. 2023, Thm. 7).

Implicit weights

Definition. For an estimator that is differentiable in the outcomes, the implicit weights are $w_i = n \, \partial\hat{\theta}/\partial Y_i$, computed for a network by one backward pass. For estimators linear in the outcomes they are the implied weights of Chattopadhyay and Zubizarreta (2023) and the outcome weights of Knaus (2024), and for posterior means the locally equivalent weights of Giordano et al. (2026), who read covariate balance as the first-order response of the estimate to an outcome shift.

Proposed result. For any such estimator, in expectation over the sampling law and under a domination condition on the derivative of $\delta \mapsto \hat{\theta}(Z_{1:n}, Y_{1:n} + \delta h(Z_{1:n}))$,

$$\partial_\delta b_n(0; h) = \mathbb{E}_P\Big[\frac{1}{n}\sum_{i=1}^{n} w_i \, h(Z_i)\Big] - \langle \alpha_0, h \rangle.$$

The estimator is first-order insensitive in the direction h exactly when its implicit weights balance h in expectation, and over a class $\mathcal{H}$ exactly when they act in expectation as the Riesz representer on $\mathcal{H}$.

Proposed result. For a corrected plug-in estimator $\hat{\theta} = n^{-1}\sum_i [\nu(W_i; \hat{g}) + \hat{a}(Z_i)\{Y_i - \hat{g}(Z_i)\}]$ with a linear smoother $\hat{g} = \hat{S}Y'$ and correction weights $\hat{a}$ fitted on an independent sample, $\partial_\delta b_n(0; h) = \mathbb{E}\langle \alpha_0 - \hat{a}, \, \hat{S}h - h \rangle$, where $\hat{S}h$ is the smoother applied to h. The sensitivity of the

AIPW estimator is therefore bounded by the product of the two errors, and that of the plug-in estimator, with $\hat{a} = 0$, is of first order in the error of the smoother.

When ν does not depend on P, as for the ATE, the sample imbalance $\hat{\Delta}(h) = n^{-1} \sum_i w_i h(Z_i) - n^{-1} \sum_i \nu(W_i; h)$ can be computed without g_0 or α_0, for instance in the direction of an estimated treatment regression $\hat{m}(X)$.

Proposed results for in-context learners

- Bayes rule of the label correction. Since the Bayesian bootstrap weights have mean $1/n$, the posterior mean under the label correction equals the Bayes rule for the one-step label $n^{-1} \sum_i [\tau_\xi(X_i) + \alpha_\xi(Z_i)\{Y_i - g_\xi(Z_i)\}]$ of Z. Zhang (2026), with $\tau_\xi = g_\xi(1, \cdot) - g_\xi(0, \cdot)$. This Bayes rule is the posterior average of the AIPW estimator, which Antonelli, Papadogeorgou, and Dominici (2022) propose. Under independent mechanisms it equals the AIPW estimator at the posterior means $\hat{\tau}_\Pi$, $\hat{g}_\Pi$ and $\hat{\alpha}_\Pi$ of τ_ξ, g_ξ and α_ξ, and when ξ_D and (ξ_X, ξ_Y) are dependent under Π it differs from it by $-n^{-1} \sum_i \mathrm{Cov}_\Pi\{\alpha_\xi(Z_i), g_\xi(Z_i) \mid \mathcal{D}_n\}$. The nuisances are evaluated on the rows to which they were fitted, so the empirical process term has to be controlled without sample splitting (Chen, Syrgkanis, and Austern 2022), and Melnychuk, Javurek, and Feuerriegel (2026) assume a stochastic equicontinuity condition that makes it negligible.
- Finite-sample analysis of the prior correction. In the Gaussian partially linear model $D = X\gamma + V$, $Y = \theta D + X\beta + \varepsilon$ with known γ, take a flat prior on θ and $\beta = \tilde{\beta} + \kappa\gamma$ with a Gaussian prior on $\tilde{\beta}$ and a flat prior on κ, which under the flat prior on θ is the correction of the outcome regression along α_ξ (Linero 2021, 2024). The posterior mean of θ is then unbiased given X for every β whenever this conditional mean exists. The correction is specific to one functional (Ritov et al. 2014, Thm. 4.3).
- Cross-fitting in context. Predicting fold I_k as a query set with the other folds as context gives an exact cross-fit with K forward passes when query rows do not influence the representations of the context rows.
- Score layers. Y. Wang, Secrest, and Tchakoute (2026) train a propensity head jointly with CausalPFN and apply AIPW and TMLE outside the network. An AIPW layer inside the network on nuisance heads that predict g and α attains the Bayes rule with $\hat{\alpha} \equiv 0$ under an end-to-end loss on θ_ξ, so its heads need not output the nuisances. The orthogonal structure is kept when the heads are trained on their nuisance targets and the gradient of the loss on θ_ξ is stopped at the heads.
- Riesz representers in context. A learner trained on the exact α_ξ or with the Riesz loss outputs $\hat{\alpha}$ for any linear functional without inverting a propensity score, which extends the propensity head of Y. Wang, Secrest, and Tchakoute (2026).

Simulation in a Gaussian partially linear model

In a simulation of posterior means, computed in closed form or by Gibbs sampling, let $D = X\gamma + V$ and $Y = \theta D + X\beta + \varepsilon$ with standard Gaussian noise, $p = 100$ covariates, $\theta_0 = 1$ and $\beta = \gamma = u_1$ for a unit vector u_1, so the outcome and the treatment depend on the same covariate direction. The prior with independent mechanisms puts a flat prior on θ and independent priors $\beta \sim N(0, I/p)$ and $\gamma \sim N(0, I/p)$, and its posterior mean is $\hat{\theta}_{\mathrm{I}} = D^\top \mathsf{M}_\lambda Y / D^\top \mathsf{M}_\lambda D$ with $\mathsf{M}_\lambda = I - X(X^\top X + \lambda I)^{-1} X^\top$ and $\lambda = p$. The conditional bias $D^\top \mathsf{M}_\lambda X\beta / D^\top \mathsf{M}_\lambda D$ of $\hat{\theta}_{\mathrm{I}}$ (P. R. Hahn et al. 2018, eq. 2.3) is the imbalance $n^{-1}\sum_i w_i (X\beta)_i - \langle \alpha_0, X\beta \rangle$ of its implicit weights on $X\beta$, since $\langle \alpha_0, X\beta \rangle = 0$. Under the corrected prior, β is replaced by $\beta + \kappa\gamma$ with $\kappa \sim N(0, 10^2)$ and γ unknown. Since $\alpha_\xi = D - X\gamma$ under unit noise variance and the prior on θ is flat, this is the correction of the outcome regression along α_ξ.

Estimator	$n =$ 200		$n =$ 800		$n =$ 3200	
	bias/sd	cover	bias/sd	cover	bias/sd	cover
least squares	−0.00	97.6	−0.06	94.2	+0.03	95.8
posterior mean, independent mechanisms	+4.81	0.4	+3.17	12.2	+1.72	60.4
one-step correction of the posterior mean	+2.81	8.6	+0.64	84.2	+0.13	93.4
DML, cross-fitted ridge	+1.43	50.0	+0.15	92.2	+0.06	93.2
posterior mean with covariate $X\hat{\gamma}$	+0.02	96.4	−0.06	94.2	+0.03	95.8
posterior mean, corrected prior	+0.99	89.0	+0.22	93.0	+0.08	93.5

Bias in units of the sampling standard deviation and coverage (%) of nominal 95% intervals, 500 replications and 200 for the corrected prior. At truths with β orthogonal to γ and at truths drawn from the prior, the posterior mean under independent mechanisms covers between 94.8% and 96.6% and has the smallest root mean squared error of the estimators compared.

Work plan, first evidence and risks

Work packages.

1. Theory of the criterion, its relation to regularity and to orthogonal scores, conditions under which the implicit weights concentrate, and the imbalance in the estimated direction $X\hat{\gamma}$.
2. Bayes rules for training labels and priors beyond the Gaussian model, including the empirical process term without sample splitting.

3. Experiments that compute implicit weights and sensitivities of published causal PFNs and of networks with the label and prior corrections on semi-synthetic data with aligned confounding, compared with DML with in-context nuisances and cross-fitting.
4. Designs with score layers that stop the gradient at the nuisance heads and with Riesz heads in context, assessed with the diagnostic.

First evidence. In the simulation in the Gaussian partially linear model with $p = 100$ and $n = 200$, the imbalance of the implicit weights in the treatment direction $X\gamma$, averaged over 100 replications, orders the five estimators by their bias. It is 0.30 for the posterior mean under independent mechanisms (bias 4.8 sd), 0.19 for its one-step correction (2.8 sd), 0.14 for DML with cross-fitted ridge (1.4 sd), -0.003 for the posterior mean with the covariate $X\hat{\gamma}$ (0.02 sd) and 0 for least squares (0.00 sd).

Risks.

- If the implicit weights of a network that is nonlinear in the outcomes do not concentrate, the diagnostic is descriptive.
- Outcomes enter a PFN after standardisation, and several PFNs return binned predictive distributions, so the derivative with respect to Y_i has to be taken through these steps.
- Work on causal PFNs appears quickly, as Melnychuk, Javurek, and Feuerriegel (2026) and Y. Wang, Secrest, and Tchakoute (2026) show, so the scope of the project has to be fixed early.

Further research questions

	Question	Closest prior work
1	Concentration of the implicit weights of nonlinear estimators, so that the sample imbalance gives a test with a known level	(Giordano et al. 2026; Chen, Syrgkanis, and Austern 2022)
2	First-order insensitivity of trained causal PFNs, with and without the label and prior corrections, measured through implicit weights	(Melnychuk, Javurek, and Feuerriegel 2026; Melnychuk, Balazadeh, et al. 2026)
3	Finite-sample properties of the prior correction of the outcome regression beyond the Gaussian partially linear model	(Melnychuk, Javurek, and Feuerriegel 2026; Linero 2024)
4	Sensitivity along shifts of the treatment mechanism and the covariate law, and a characterisation of the estimators that are first-order insensitive in all directions through the gradient of θ	(Chernozhukov, Escanciano, et al. 2022)

	Question	Closest prior work
5	An in-context learner that outputs the Riesz representer of a functional given as input	(Y. Wang, Secrest, and Tchakoute 2026; J. Zhang et al. 2024; Chernozhukov, Newey, Quintas-Martínez, et al. 2022)
6	Compositions of learned components, and conditions under which learned representations span both g_0 and α_0	(Melnychuk, Frauen, et al. 2026; Belloni, Chernozhukov, and Hansen 2014; Chernozhukov, Newey, Singh, et al. 2022)
7	Leave-one-out stability of attention as a replacement for sample splitting	(Nagler 2023; Chen, Syrgkanis, and Austern 2022)